

How IrDA Works

System support

To communicate with an IrDA enabled device, your computer needs to have an IrDA port. Some motherboard models come with an infra-red port on board, but most include only the IrDA header, which is a connector between the motherboard and the back panel of the cabinet. The IrDA adapter which contains the infra-red transmitter and receiver has to be connected to the IrDA port at the rear of the cabinet. Usually, motherboard manufacturers don't include the IrDA adapter, even if the motherboard has an IrDA port. Read your motherboard manual to find out if you have an IrDA port. If you do, buy an adapter from your dealer.

Another option is to use USB stick-based IrDA ports. Just plug one of these devices into a vacant USB port, install the drivers if required, and your computer is IrDA-ready. These devices are available for about Rs 1,000.

Pros

Being the simplest of the lot, it really doesn't make sense to buy a product for its IrDA unless you have an existing IrDA-enabled device or system. It is best suited for transferring small amounts of data such as schedules, contacts, etc. Use it for transferring data from your system to a mobile device or between two mobile devices.

Cons

Don't use IrDA for transferring large amounts of data between a laptop and a system. Although data can be transferred, the slow data rate and line of sight requirement will make it a very avoidable experience.

Devices using infra-red

IrDA ports are found in personal electronics devices such as mobile phones, PDAs, remote controls, etc. Nokia models such as the 6510, 8250, the Sony Ericsson T310 and T610, the Palm Zire 71, etc., all come with infra-red ports.

If your mobile phone is IrDA-capable, you can transfer ring tones, wallpapers, contacts, etc., to and from your phone. However, you will need to have a mobile phone management utility such as Oxygen Phone manager. Get it at www.oxygenphonemanager.com. Similarly, you can transfer data and synchronize your PDA with your computer.

Interestingly, IrDA-enabled PDAs can be used as substitutes for remote controls. All you need to do is load an application such as OmniRemote (www.pacificneotek.com) onto your PDA, point a remote towards the PDA and press the most commonly used keys. For a television remote, you probably use Channel up, Channel down, Volume up and Volume down more than any other buttons. After you press a key on the remote, you can assign a PDA key to mimic the function. Once you've assigned the keys, you're all set to use your PDA as a remote—secretly switching channels while your father sits guarding the remote, gaping at the swimsuit round of Miss World, is one sure way of getting a few laughs.

look for IrDA-enabled devices through a process called discovery. Once it finds a secondary device, two random 32-bit addresses are generated for each device—even if the same address is generated for both devices, there are mechanisms to resolve them. Once the addresses are assigned, a connection is established and the devices figure out each other's capabilities: supported data rates, data size, turnaround time, etc. After these rituals, the data transfer commences at the highest supported speed. A 'permission to transmit token' is used, which ensures only one device talks at a time. Only the device possessing the token can transmit data. Though more than two devices can communicate with each other, only one can talk.

There is no fixed speed for IrDA devices, but all devices need to be capable of at least 9.6 Kbps, as this is the speed at which initial discovery and negotiation takes place. Currently, the maximum supported speed is 4 Mbps, but a 16 Mbps standard is in the pipeline. IrDA devices can be up to a metre apart, but work best within a range of 5 cm to 60 cm. As IrDA commu-

nication happens only by conscious effort of a user, the range is short and security isn't considered an issue—there isn't any in the standard.

Bluetooth

Bluetooth is designed to be a personal area network, where participating entities are mobile and require sporadic communication with others. It's omni directional; i.e., it doesn't have the line of sight limitation like infra-red does. Ericsson started the work on Bluetooth, and being a Denmark-based company, they named it after the Danish king Harold Bluetooth. Later, a Bluetooth special interest group, comprising companies such as Nokia and Ericsson, was formed to oversee the development and standardisation of the technology.

Bluetooth operates in the 2.4 GHz area of the spectrum, and provides a range of 10 metres. It offers transfer speeds of around 720 Kbps. When

Bluetooth devices are connected to each other, they form a network called piconet. Each piconet can have a master, and up to seven slaves. Communication takes place only between the master and the slave. If the slaves need to communicate with each other, they can't route via the master—they have to form a separate piconet. A device can participate in more than one piconet at any given time—it can be a slave in one piconet, and a master in another. Such networks are called scatternets, where different piconets are connected using a common device.

To accommodate interference from other signals, Bluetooth uses frequency



Not just mobile phones and PDAs, Bluetooth can be used for other wireless applications like this Microsoft Bluetooth keyboard and mouse